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EDUCATION

NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE Aug 2024 - Dec 2027 expected

B.Sc. in Mathematics | School of Physical and Mathematical Sciences | Singapore

- Current CGPA: 4.50/5.00.
- Coursework includes linear algebra, probability and statistics, stochastic processes, regression analysis, time series analysis, machine learning, reinforcement learning, database design, and visual and predictive analytics.

University of California, Berkeley Jan 2027 - May 2027 planned

University of California Education Abroad Program exchange

- Planned exchange focused on mathematics; academic interests include partial differential equations, optimization, mathematical economics, numerical analysis, financial engineering, and quantum algorithms.

PUBLICATION

MLP-Initialized Reinforcement Learning for Exit Timing in SPX Short-Straddle Portfolios 2026

Jianning Wang, Robert Ślepaczuk, and Paweł Sakowski

- Accepted at the 7th ACM International Conference on AI in Finance (ICAIF 2026).
- arXiv:2610.13126 | ACM DOI: 10.1145/3858063.3858801

RESEARCH EXPERIENCE

University of Warsaw: Visiting Researcher May 2026 - Aug 2026

- Developed an MLP-initialized reinforcement learning framework for exit timing in SPX short-straddle portfolios, accounting for transaction costs and realistic execution constraints.
- Built an end-to-end Python research pipeline covering oracle-label construction, supervised MLP pretraining, reinforcement learning fine-tuning, and portfolio backtesting.
- Implemented a rolling portfolio simulation with margin-based sizing and liquidation logic; evaluated out-of-sample NAV, drawdown, and risk-adjusted performance against hold-to-expiry and other exit rules.
- Co-authored the resulting paper accepted at ACM ICAIF 2026.

Nanyang Technological University Aug 2025 - Jul 2026

Research Project | Regime-aware reinforcement learning for quantitative trading

- Designed a regime-aware Mixture-of-Experts framework combining Hidden Markov Models (HMMs) and multiple reinforcement learning agents for adaptive trading strategy selection.
- Used technical features from DJI30 constituents to identify latent market regimes and condition trading policies on inferred states.
- Evaluated PPO, A2C, DDPG, TD3, and SAC under portfolio constraints, examining policy behavior across changing market conditions.

Bayesian Volatility Estimation for Black-Scholes Option Pricing Apr 2025 - Aug 2025

Research Project

- Applied Bayesian updating to estimate the volatility parameter in Black-Scholes option pricing using market observations.
- Implemented posterior inference with PyMC3 and MCMC sampling; developed a modular Python workflow using NumPy, Pandas, SciPy, and Matplotlib.
- Compared the approach with stochastic-volatility and local-volatility benchmarks, including Heston and Dupire formulations.

INDUSTRY EXPERIENCE

Beijing Gold Prospecting Technology Ltd.: Quantitative Research Intern Nov 2025 - Feb 2026

- Built a Random Forest alpha model for CSI500/CSI1000 with rolling five-year training windows and weekly signal generation. Model evaluation reported approximately 30% higher Sharpe ratio and 10% higher annualized return than the legacy linear model.
- Developed an end-to-end data pipeline integrating market data, fundamentals, and index weights, with incremental retrieval, trading-calendar alignment, ticker matching, and missing-value handling.
- Applied Bayesian optimization to hyperparameter tuning and feature selection; mined and evaluated candidate factors for predictive power and out-of-sample portfolio contribution.
- Cached daily API data and automated standardized holdings-signal exports, synchronizing trading dates across the signal-to-execution workflow.

China Huadian Corporation: Investment Analyst Intern Apr 2025 - Jul 2025

- Developed a financial forecasting model for a 347 MW distributed photovoltaic project; conducted IRR and payback-period sensitivity analyses to assess project economics.
- Modeled capital structures for a new energy fund, including preferred/subordinated capital and hybrid investment structures; contributed to exit scenarios covering IPOs, M&A, and REITs.
- Supported partnership due diligence and developed a risk-assessment framework linking investment risks, return distributions, and potential exit pathways.

ENTREPRENEURSHIP AND LEADERSHIP

Beishan Education Technology Co., Ltd.: Founder Jun 2024 - Present

- Founded an education business providing education technology, admissions consulting, and tutoring services; current annual revenue is approximately RMB 500,000.
- Coordinated product, operations, and student services; managed day-to-day work across a team of more than 10 and served 15 families in the first year.
- Established service protocols and performance-monitoring processes to improve consistency and support referrals and business growth.
- Used R to track business profits and employee compensation for financial monitoring and operational decisions.

PROJECTS AND COMPETITIONS

QAI Ventures Hackathon Oct 2025

Top 10 Finalist | Quantum autoencoder for credit-card fraud detection

- Developed an unsupervised anomaly-detection pipeline using a four-qubit Quantum Autoencoder (QAE); applied PCA followed by amplitude encoding to represent transaction features.
- Trained on normal transactions and used reconstruction fidelity as an anomaly score; achieved an evaluation AUC of 0.9456.
- Built a modular, reproducible workflow with training-loss curves and ROC visualizations.

Statistical Arbitrage in MA-EG Futures Oct 2025 - Nov 2025

Quantitative Research Project

- Built a backtesting framework with continuous-contract alignment, rolling-correlation filters, ATR-based risk-parity sizing, and slippage modeling.
- Designed a triple-stochastic Stress Indicator to identify extreme spread deviations and potential mean-reversion entries between MA and EG futures.
- Evaluated 70 historical trades under predefined entry and exit rules, with a reported annualized return of 12.71% in backtesting.

Misinformation Detection with Machine Learning Sep 2025 - Oct 2025

Machine Learning Project

- Developed and compared Logistic Regression and CART models; explored XGBoost as an additional model.
- Prepared more than 30 features, including source reliability, fact-checking signals, and platform/country metadata, using one-hot encoding, normalization, and stratified data splitting.
- Obtained approximately 86% accuracy for Logistic Regression and CART; evaluated confusion matrices and false-positive/false-negative rates.

Faculty Salary and Gender Pay Analysis Aug 2025 - Sep 2025

Statistical Analysis Project

- Used R and multiple linear regression to investigate the association between gender and faculty salaries, controlling for experience, rank, certification, publication rate, and clinical emphasis.
- Conducted VIF diagnostics, AIC comparisons, and stepwise model selection; produced more than 15 visualizations of salary structure and model results.

PyImageStudio Image Editing Application Oct 2024 - Dec 2024

Python Development Project | Nanyang Technological University

- Developed a modular NumPy-based image editor with filters for brightness, contrast, blur, edge detection, and embossing, including convolution-kernel implementations.
- Implemented a Magic Wand tool with breadth-first flood fill and Delta-E color distance; designed mask-based, non-destructive editing for region-specific effects.

ADDITIONAL RESEARCH AND INVESTMENT PROJECTS

Bayesian Reinforcement Learning and Monte Carlo Tree Search 2025

Sequential Decision-Making Research Project

- Designed a policy-optimization framework combining Bayesian Reinforcement Learning (BRL) and Monte Carlo Tree Search (MCTS) for sequential decisions under uncertainty.
- Explored algorithmic trading and online bidding as application settings for adaptive policy updates.

Self-Directed U.S. Equity Portfolio 2025

Personal Investment Project

- Managed a personal U.S. equity portfolio focused on large-cap technology stocks, including NVIDIA, Alphabet, Meta, Microsoft, and Tesla.

SKILLS AND RESEARCH INTERESTS

Technical Skills

- Programming and tools: Python, SQL, R, C++, Git, LaTeX, Microsoft Office, and MongoDB.
- Libraries and methods: NumPy, Pandas, SciPy, Matplotlib, PyTorch, TensorFlow, PyMC3, Random Forests, Bayesian optimization, HMMs, and reinforcement learning.
- Quantitative research: feature engineering, alpha-factor research, portfolio backtesting, risk modeling, time series, and derivatives.
- Research interests: financial mathematics, machine learning, reinforcement learning, and quantum algorithms.
- Languages: Mandarin Chinese and English; basic French (A2) and German (A1).

SPORTING BACKGROUND

Alpine Skiing and Football

- Former alpine skiing athlete in Miyun District, Beijing; former youth player at the Beijing Guoan Football Club academy.